SLIDE 1 OF 5: EXECUTIVE OVERVIEW & CERL MISSION ALIGNMENT

- Solution: Project Ebony v8.0.0: Air-Gapped Tri-Brain OT/SCADA Safety Interlock, Automated Bridge/Pavement NDE & WORM Merkle Provenance Testbed
- Offeror: Humphrey Virtual Farms LLC (CAGE: 1AHA8 | UEI: S1M4ENLHTDH5 | Madill, OK 73446)
- Status: Non-Traditional Defense Contractor (NTDC) & Small Disadvantaged Business (SDB)
- CERL Alignment: Operational Energy (Microgrids/Switchgear), Sustainment Management System (SMS/BUILDER Sensor Analytics), Robotics for Engineering Operations (Autonomous NDE)
- Verified Federal Docket: DoD CDAO Tradewinds Docket 10-26-4046 (Status: Received | 04:24 Video SHA-256: c597865785eae6fdad6fdf6803a2710aaa0fb5c2fa28fe512b6ed76ef8b7a7fd)

SLIDE 2 OF 5: THE <4.16ms INSTALLATION OT/SCADA & SUSTAINMENT PHYSICS GAP

- GAO-21-81 & GAO-24-106137: DoD operates 500+ installations, 300,000+ buildings, and $4.2B+ in OT energy systems on legacy PLCs scanning at 50 to 200ms.
- CISA AA23-144A / AA24-038A (Volt Typhoon) & TRISIS: Adversaries corrupt PLC safety logic while spoofing operator displays.
- Physics Gap: High-voltage arc-flash faults destroy switchgear in <4.16ms and thermal excursions occur in 6.0ms?meaning 50-200ms PLCs react 8x to 33x too late.
- Sustainment Gap: Manual bridge, pavement, and facility inspections cost $4,600+/day with snooper-truck closures and lack cryptographic chain-of-custody verification.

SLIDE 3 OF 5: LIVE TRL 4 AIR-GAPPED TRI-BRAIN ARCHITECTURE (127.0.0.1 LOOPBACK)

- Baseline Maturity: TRL 4 laboratory software emulation testbed (0% cloud, 0 external retrieval APIs, 0 open WAN ports, 637 C2 Python modules, 85/85 passing unit test suites, 17/17 blocked Red-Team vectors, 52 bare-metal SQLite WORM tables).
- Brain 1 (Safety Interlock): Models <50us Cortex-M7/RISC-V bare-metal loop, 2.04us RS-485 Modbus FC05 trip, and 36.75ns SiC MOSFET cutoff target at 28.5W on 24V DC.
- Brain 2 (Isolated NDE): Read-only buffer (0 actuator wire) executing offline ASTM D6433 PCI and structural crack mensuration ($250/sortie vs. $4,600/day manual).
- Brain 3 (WORM Provenance): Seals SHA-256 Merkle DAG (Root:

bb39eeaa66a39ad2e0c0ab74ba4a5c4faac2bf0d98babc0362482b97607151a1) in <0.15ms.

SLIDE 4 OF 5: 5-VENDOR QUANTITATIVE ANALYSIS OF ALTERNATIVES (6/6 GATES PASSED)

- Project Ebony v8.0.0 (CAGE 1AHA8): 6/6 Gates Passed | Modeled 36.75ns cutoff (1,360,544x faster than 50ms PLC) | 28.5W target draw | 100% air-gapped SQLite WORM.
- Siemens S7-1500 / Rockwell ControlLogix: 0/6 Gates | 50-200ms scan | 150-450W draw.
- Schneider Triconex SIS: 0/6 Gates | 10-50ms scan | 200-600W draw | Breached by TRISIS.
- Dragos / Nozomi Passive IDS: 0/6 Gates | 0 physical cutoff authority | 300-1,200W draw.
- Cloud AI Wrappers (Jev AI): 0/6 Gates | >250ms cloud RTT | Fails under EMCON.

SLIDE 5 OF 5: TRACTION, DUAL-USE CLIN PRICING & 12-MONTH PHASE-1 PROTOTYPE PLAN

- Verified Traction: Active CDAO Tradewinds Docket 10-26-4046; USACE ERDC BAA filings (CERL, ITL-4, GSL-19); 11-artifact Microgrid SOW suite; 110-control NIST 800-171/800-82 map.
- Phase-1 Prototype ROM: $1,200,000.00 (12 Months; TRL 4 to TRL 6; $150,000.00 Mobilization) or 26-Module Commercial CLINs (CLIN 0001 $2,450/mo; CLIN 1004-A $895K/yr).